In the right angled ΔBAC, $\tan 30° = \dfrac{AB}{AC}$

$$\frac{1}{\sqrt{3}} = \frac{200}{AC} \Rightarrow AC = 200\sqrt{3} \quad \ldots(1)$$

In the right angled ΔBAD, $\tan 45° = \dfrac{AB}{AD}$

$$1 = \frac{200}{AD} \Rightarrow AD = 200 \quad \ldots(2)$$

Now, $CD = AC + AD = 200\sqrt{3} + 200$ [by (1) and (2)]

$CD = 200(\sqrt{3}+1) = 200 \times 2.732 = 546.4$

Distance between two ships is 546.4 m.

Example 6.22 From a point on the ground, the angles of elevation of the bottom and top of a tower fixed at the top of a 30 m high building are $45°$ and $60°$ respectively. Find the height of the tower. $(\sqrt{3} = 1.732)$

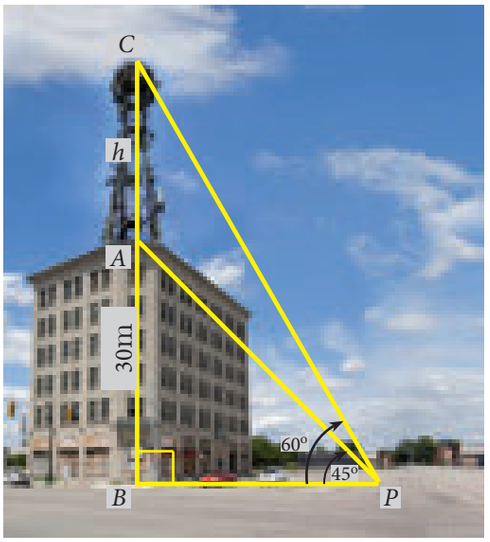


Fig. 6.16

Solution Let AC be the height of the tower.

Let AB be the height of the building.

Then, $AC = h$ metres, $AB = 30$ m

In the right angled ΔCBP, $\angle CPB = 60°$

$$\tan\theta = \frac{BC}{BP}$$

$$\tan 60° = \frac{AB + AC}{BP} \Rightarrow \sqrt{3} = \frac{30 + h}{BP} \quad \ldots(1)$$

In the right angled ΔABP, $\angle APB = 45°$

$$\tan\theta = \frac{AB}{BP}$$

$$\tan 45° = \frac{30}{BP} \Rightarrow BP = 30 \quad \ldots(2)$$

Substituting (2) in (1), we get $\sqrt{3} = \dfrac{30+h}{30}$

$h = 30(\sqrt{3} - 1) = 30(1.732 - 1) = 30(0.732) = 21.96$

Hence, the height of the tower is 21.96 m.

Example 6.23 A TV tower stands vertically on a bank of a canal. The tower is watched from a point on the other bank directly opposite to it. The angle of elevation of the top of the tower is $58°$. From another point 20 m away from this point on the line joining this point to the foot of the tower, the angle of elevation of the top of the tower is $30°$. Find the height of the tower and the width of the canal. $(\tan 58° = 1.6003)$

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